

KUNAL ROY CHOWDHURY

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SUMMARY

Business Analytics professional with 2+ years of quantitative consulting experience at Protiviti, ranked in the top 5% of 3,500+ global consultants for delivering data-driven risk frameworks and statistical insights to Fortune 10 clients. Currently advancing expertise in Machine Learning, Predictive Modeling, and Generative AI through an MS in Business Analytics & AI at the University of Maryland, Smith School of Business. Combines technical proficiency (Python, R, SQL, Power BI, Tableau) with domain expertise in Enterprise Risk, SOX/FCPA compliance, and analytical storytelling. Published researcher in AI infrastructure strategy. Trilingual: English, Hindi, Bengali. Seeking analytical roles in Strategy, Data Science, or Enterprise Risk where quantitative insights drive measurable organizational outcomes.

EXPERIENCE

Zochem LLC

Jun 2026 – Present

Technical Process Intern

- Mapped zinc oxide manufacturing workflows to identify key operational weaknesses, translating data-driven findings into targeted process improvement recommendations that strengthened production efficiency across multiple production lines.
- Analyzed shift-level output and production bottleneck metrics, collaborating with cross-functional teams to implement corrective actions and drive measurable improvements in throughput consistency and operational reliability.

University of Maryland, College Park

Mar 2026 – May 2026

Research Assistant

- Tracked registration and engagement metrics for the 2026 UMD Agentic AI Challenge, driving a 40% increase in student participation through data-informed advertising and outreach strategies.
- Collaborated with UMD's Division of Information Technology and industry partners to conduct structural analysis and refinement of real-world AI case models, improving statistical robustness and applicability of competition evaluation frameworks.

Protiviti

Jan 2025 – Jun 2025

Consultant, Enterprise Risk & Strategy | ★ Quarterly Rising Accelerator Award — Top 5% of 3,500+ Consultants

- Led a team of 4 junior consultants to design and refine ERM frameworks for Fortune-level clients, embedding quantitative risk analysis and data-driven decision criteria into clients' strategic planning processes.
- Engineered a real-time project tracking and data visibility platform on Azure DevOps, consolidating multi-workstream metrics and reducing status reporting latency across concurrent client engagements.
- Identified 30+ process inefficiencies through quantitative workflow analysis across client operations, delivering data-driven remediation strategies that improved operating effectiveness by 30%.
- Prepared and presented board-ready statistical risk reports to C-suite leadership, translating complex analytical findings into quantified, actionable executive recommendations.

Protiviti

Sep 2023 – Dec 2024

Junior Consultant, Risk & Compliance Analytics

- Designed and statistically tested 70+ internal business controls for a Fortune 10 technology client, verifying regulatory compliance with FCPA, SOX, and Partner Agreement standards through comprehensive data validation.
- Sustained a 97% on-time project completion rate across international engagements by partnering with cross-functional stakeholders to optimize business processes and KPI tracking frameworks.
- Delivered data-driven corrective action frameworks and audit reports that quantified revenue potential and statistically mitigated regulatory risk exposure for senior management.

SKILLS

Analytics & Data Science: Python, R, SQL, Machine Learning (Random Forest, XGBoost, K-Means, SMOTE, NLP/TF-IDF), Predictive Modeling, Statistical Analysis, Generative AI

Business Intelligence & Visualization: Power BI, Tableau, Microsoft Excel (Advanced), Azure DevOps, Data Visualization, Executive Reporting

Risk & Strategy: Enterprise Risk Management (ERM), SOX Compliance, FCPA, Business Analysis, Process Optimization, Strategic Planning, KPI Development

Languages: English (Native) | Hindi (Native) | Bengali (Advanced)

EDUCATION

University of Maryland, Smith School of Business

Aug 2025 – Dec 2026

MS, Business Analytics & Artificial Intelligence

Terrapin Scholar | GRE 325 | President, MS Business Analytics Association

Indian Institute of Technology, Guwahati

Jan 2025 – Jul 2025

Certification in Data Analytics with Generative AI

Guru Gobind Singh Indraprastha University, New Delhi

Aug 2019 – Jun 2022

BBA, Business Administration

KEY ACHIEVEMENTS

Quarterly Rising Accelerator Award

Ranked in top 5% of 3,500+ global Protiviti consultants for exceptional project delivery and measurable client impact.

Published Author — [India Strategic Journal](#)

Conducted extensive secondary research and statistical synthesis on domestic AI infrastructure development in India; authored peer-reviewed strategic position piece titled Betting Big on AI analyzing corporate drivers of technological advancement and implications for national competitiveness

Terrapin Scholar

Merit-based scholarship awarded by the University of Maryland, Smith School of Business, recognizing prior academic and professional excellence.

ACADEMIC PROJECTS

Yelp Review Usefulness Prediction — Python, NLP, XGBoost, TF-IDF, Sentiment Analysis

Spring 2026

- Built an end-to-end NLP pipeline combining TF-IDF text vectorization with user engagement metadata to predict whether a Yelp review would be marked useful, outperforming baseline models across a large-scale real-world dataset.
- Achieved XGBoost early stopping at round 387 with Training AUC of 0.930 and Validation AUC of 0.904, demonstrating strong generalization with minimal overfitting.
- Engineered 15+ features spanning review length, user activity history, and temporal metadata; applied SHAP value analysis confirming review length and elite user status as the strongest predictors of usefulness.

CAVA Group: National Growth Strategy — Python, Excel, Supply Chain Optimization

Spring 2026

- Developed a data-driven 3–5 year national expansion strategy using Excel-based demographic density modeling, identifying Texas and Florida as high-potential markets where household income exceeded state medians by 11–24%.
- Constructed a Python supply chain optimization model for a Sustainable Moat vertical integration strategy, enabling a 14% pricing advantage relative to industry inflation and strengthening long-term competitive positioning.
- Benchmarked CAVA against Chipotle and Sweetgreen using comparative financial and operational metrics, positioning CAVA's value-for-money offering as the primary customer acquisition lever in high-growth suburban corridors.

Obesity Risk Prediction & Lifestyle Analytics — Python, Scikit-Learn, Random Forest, SMOTE, K-Means

Fall 2025

- Built a multi-class Random Forest classification engine with SMOTE-based class imbalance correction, achieving 95.3% predictive accuracy and 98% recall on high-risk patient segments across 2,111 patient profiles.
- Applied unsupervised K-Means clustering (Silhouette Score: 0.42) to segment patients by behavioral risk factors, identifying Vegetable Consumption and Meal Frequency as the top predictive variables via feature importance analysis.
- Validated model on a holdout test set (n=422), achieving 94.8% accuracy, and produced risk-tier classifications enabling healthcare providers to prioritize high-risk patient outreach at scale.

NYC Taxi Demand Forecasting — Python, Time Series, XGBoost, Feature Engineering

Fall 2025

- Developed a borough-level demand forecasting model for NYC yellow cab trips using 3+ years of TLC trip record data, engineering time-based features (hour, weekday, holiday flags) to capture cyclical demand patterns.
- Trained and compared ARIMA, Random Forest, and XGBoost regressors, achieving a 12.4% improvement in RMSE over baseline with XGBoost after systematic hyperparameter tuning via cross-validated grid search.
- Visualized demand heatmaps and temporal trend decompositions to surface actionable insights for fleet reallocation, identifying peak demand windows with 89% precision across 5 borough clusters.